MIL-P-60090B(EA)
3 December 1980
SUPERSEDING
MIL-P-60090A(MU)
5 July 1968

MILITARY SPECIFICATION

PROJECTILE, 105MM, WP-T, M416;

FILLING ASSEMBLY

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the white phosphorus (WP) filling assembly of the M416 projectile, without fuzing components.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-215 - Chemical Agent, White Phosphorus (WP).
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specifications For.

FSC 1315

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-453 - Radiographic Inspection.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

8886485 - Projectile, 105MM, WP-T, M416; Filling Assembly.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Dwg 8886485 and associated drawings, and with all requirements specified in the applicable specifications and standards.

3.3 Radiographic quality Radiographs for determination of the filling height shall be taken in accordance with MIL-STD-453 and shall be of a quality which will permit proper evaluation of the filling assembly. The minimum optical density of each radiograph in the area being inspected shall not be less than 1.25 and shall be within a density range that will permit proper identification of filling height rejects.

3.4 Filling of projectile.

3.4.1 Projectile and case impeller. The projectile cavity and case impeller shall be clean and free of volatile corrosion inhibitor paper, dirt, burrs, rough edges, and foreign matter.

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3.4.2 Filling. The projectile shall be filled with molten WP conforming to MIL-C-215 at a temperature of 140° ± 10°F (60° ± 5.5°C). When water is in contact with the WP during the filling operation, the water shall be buffered by a suitable alkaline salt to maintain a pH value between 6.5 and 8.0. After the test for leakage, the WP shall be solidified while maintaining the projectile in a vertical nose-up position.

3.4.3 Filling height of WP. When radiographed in a vertical, nose-up position, the depth of the water remaining in the filled and closed projectile shall not exceed one-quarter inch (0.250 inch); and, there shall be a 3 to 5 percent void, by total volume, existing above the surface of the water over the WP in the projectile (see 6.3).

3.4.4 Cleaning after projectile closing. The exterior face of the filling assembly and the interior surface of the case impeller cavity shall be cleaned thoroughly to remove all WP contamination (see 6.4).

3.5 Leakage. When tested as specified in 4.5.2, the filling assembly shall not leak when maintained at a temperature of 160° ± 5°F (71° ± 3.0°C) for a period of 8.0 hours, minimum; or, when maintained at a temperature of 207° to 212°F (97°C to 100°C) for a period of 15 minutes, minimum.

3.6 Data requirements. The contractor shall generate data in accordance with the requirements of the data item descriptions cited in 4.4.2.1.

3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship.

3.8.1 Projectile filling assembly. The projectile filling assembly shall be free from internal and external contamination (oil, grease, etc.), chips, dents, burrs, and foreign matter that may impair further assembly and proper functioning. The cleaning method used shall not be injurious to any of the filling assemblies; nor shall the assemblies be contaminated by the cleaning agent.

3.8.2 Painting and marking. The painting and marking shall be restored by the filling contractor on any components furnished to him painted and marked, when such paintings and markings have been marred or obliterated.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be performed on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

PART DESCRIPTION	DRAWING	QUANTITY
Projectile Filling Assembly	8886485	25

4.3.2 Inspections to be performed. See MIL-A-48078 and table I specified herein, see 4.3.2.1.

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60090B(EA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.1	Projectile, 105MM, WP-T, M416 Filling Assembly	1 OF 1	8886485
			NEXT HIGHER ASSEMBLY: Not Applicable

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile, Filling Assembly (Dwg 8886485)				
	Examination for defects	25		3.2	4.4.2.4
	Radiographic Examination	25		3.4.3	4.5.1
	Leakage test	25		3.5	4.5.2

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.3.3 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of projectile filling assemblies shall contain:

a. Projectile metal parts of one lot interfix number from one manufacturer;

b. Case impeller metal parts of one lot interfix number from one manufacturer;

c. WP conforming to MIL-C-215 of one lot from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

Sampling plans. Unless otherwise specified in the classification of defects and tests tables, paragraphs 4.4.2.2 through 4.4.2.5, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, inspection Level II.

4.4.2.1 Defect reporting. A report of all critical defects found, including the number of items inspected and corrective action taken, shall be prepared monthly in accordance with Data Item Description DI-R-1721 and copies delivered as specified in the contract.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60090B(EA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Projectile, 105MM, WP-T, M416, Metal Parts, Prior to Filling	1 OF 1	10522803
			NEXT HIGHER ASSEMBLY: 8886485

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Presence of nicks, burrs, rough edges or foreign matter on press fit surface		100%	3.4.1	Visual/Manual
Minor					
201	VCI paper or foreign matter in cavity		0.65%	3.4.1	Visual
202	Evidence of poor workmanship		0.65%	3.8.1	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60090B(EA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Case Impeller, Projectile Burster, M175, Prior to Assembly to Filled Projectile		10523973
			NEXT HIGHER ASSEMBLY
			8886485

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Presence of nicks, burrs, rough edges, or foreign matter on press fit surface		100%	3.4.1	Visual/Manual
102	Lubricant missing from press fit surface		100%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60090B(EA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Projectile, 105MM, WP-T, M416 Filling Assembly				8886485
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
Critical					
1	Evidence of WP leakage			3.5	4.5.2/Visual
2	WP contamination			3.4.4	Visual
Major					
101	Projectile bourrelet diameter, max.		100%	3.2	Gage
102	Case impeller below flush with shoulder		100%	3.2	Visual/Manual
103	Depth of case impeller from rear of projectile body		0.40%	3.2	Gage
104	Temperature of molten WP improper		0.40%	3.4.2	Approved Thermometer
105	pH of contact water improper		0.40%	3.4.2	Approved pH Meter
106	Depth of water above WP in filled projectile, max.		0.40%	3.4.3	Determination/4.5.1
107	Volume of void above water in filled projectile		0.40%	3.4.3	Determination/4.5.1
Minor					
201	Protector cap missing		0.65%	3.2	Visual
202	Presence of burrs, chips, corrosion, or foreign matter		0.65%	3.8.1	Visual/Manual
203	Marking incorrect, illegible, or missing		0.65%	3.2	Visual
204	Protective finish damaged		0.65%	3.8.2	Visual
205	Evidence of poor workmanship		0.65%	3.8.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60090B(EA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8837835 (3 sheets)
4.4.2.5	Pallet				NEXT HIGHER ASSEMBLY 8886485
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Strapping missing, broken, or loose		0.65%	3.1	Visual/Manual
202	Pallet damaged		0.65%	3.1	Visual
203	Improperly packed		0.65%	3.1	Visual
204	Marking missing, illegible, or incorrect		0.65%	3.1	Visual
205	Evidence of poor workmanship		0.65%	3.8.1	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Testing. Testing is described in the first article and quality conformance inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.2. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See 6.2 herein and Section 6 of MIL-A-48078.

4.5 Methods of inspection. PRECAUTION. This specification covers sampling and testing of chemical, toxic, or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations, and procedures must be followed in handling and processing these materials.

4.5.1 Radiographic determination of filling height and percent void volume. Prior to exposure to radiographic examination, the filled projectiles selected for this test shall be temperature conditioned at 140° ± 10°F (60° ± 5°C) for 15 minutes while in the vertical, nose up position (the temperature measurements shall be made using thermocouples placed in the case impellers of at least two projectiles located near the center of the sample batch when placed in the temperature conditioned chamber). Determine the height of WP fill corresponding to the required void as follows:

(a) Measure total volume of the assembled projectile including case impeller.

(b) Calculate required void volume. (3 and 5% of (a) above.)

(c) Subtract (b) from (a) to determine liquid volume.

(d) Using liquid volume minus 1/4 inch water layer develop a standard fill height measurement that will be used to read the WP fill height on the radiographs.

NOTE: It will be necessary to perform the void/fill height determinations on several projectiles of the same lot to establish a reliable fill height measurement which will not be affected by the tolerances of the metal parts.

4.5.2 Leakage. After final cleaning, the projectiles shall be placed in an oven in a vertical, nose-up orientation, and temperature conditioned at 160° ± 5°F (71° ± 3°C) for 8.0 hours, minimum, after which they shall be examined for evidence of WP leakage. Alternately, the projectiles may be placed in a vertical, nose-end down position, temperature conditioned

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to 207°F to 212°F (97°C to 100°C) for 15.0 minutes, minimum, and then examined for WP leakage. (When using the alternate method, thermocouples shall be placed in the case impeller of at least two projectiles located in the center of the temperature conditioned chamber per batch, in order to determine the temperature of the projectile batch. When the thermocouples indicate that the required temperature range has been met, the storage time shall be said to have started.) The emission of smoke before, during, or after temperature conditioning constitutes leakage.

5. PACKAGING

5.1 Packing, level A. The filled projectiles shall be palletized for shipment in the same manner as they were received.

5.2 Marking. In addition to any special marking required by the contract or order, palletized loads shall be marked in accordance with the applicable drawing and MIL-STD-129.

6. NOTES

6.1 Ordering data.

6.1.1 Procurement requirements. See MIL-A-48078.

6.1.2 Data requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.4.2.1	Quality Inspection Defect Report	DI-R-1721

(Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.1.2.1 Delivery of data to the technical activity. The following addresses are to be added to those listed in the contract (Form DD 1423) for delivery of one copy, each, of the Ammunition Data Cards and Quality Inspection Defect Reports prepared by the contractor for the duration of the contract: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAC-D, and DRDAR-CLN-SE, Aberdeen Proving Ground, MD 21010.

6.2 Submission of inspection equipment designs for approval. Submit designs, as required, to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAC-D and DRDAR-CLN-SE, Aberdeen Proving Ground, MD 21010.

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6.3 Projectile filling (advisory). Projectiles filled with approximately 6.0 pounds of WP have been found to comply with the requirements of 3.4.3.

6.4 Decontamination (advisory). Filled and closed projectiles may be decontaminated by passing the projectile through a hot water and detergent spray washer, or by flushing with hot water while scrubbing vigorously with steel wool or a wire brush. Additional cleaning of the case impeller may be accomplished with a steam lance and a power driven wire brush.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-P-46070B

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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